Supplementary materials

Supplementary figure 1 Selection of articles for systematic review

Supplementary figure 2 The gender of screening participants

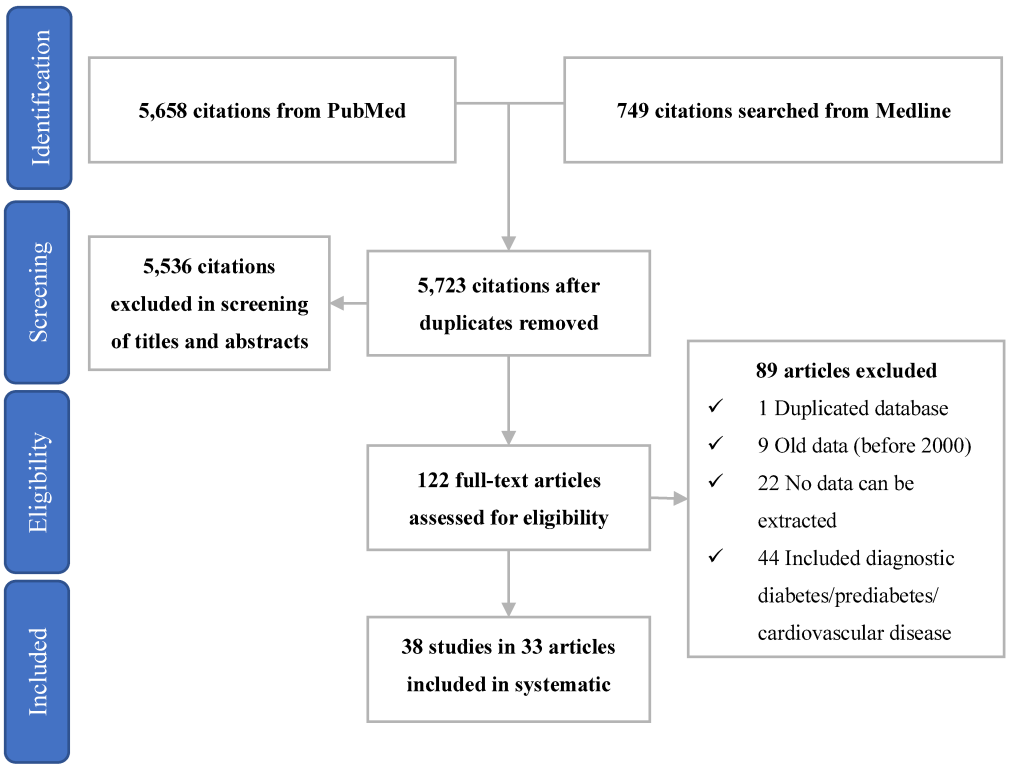
Supplementary table 1 Summary of diabetes and lipid screening guidelines

Supplementary table 2 Search strategy for article selection

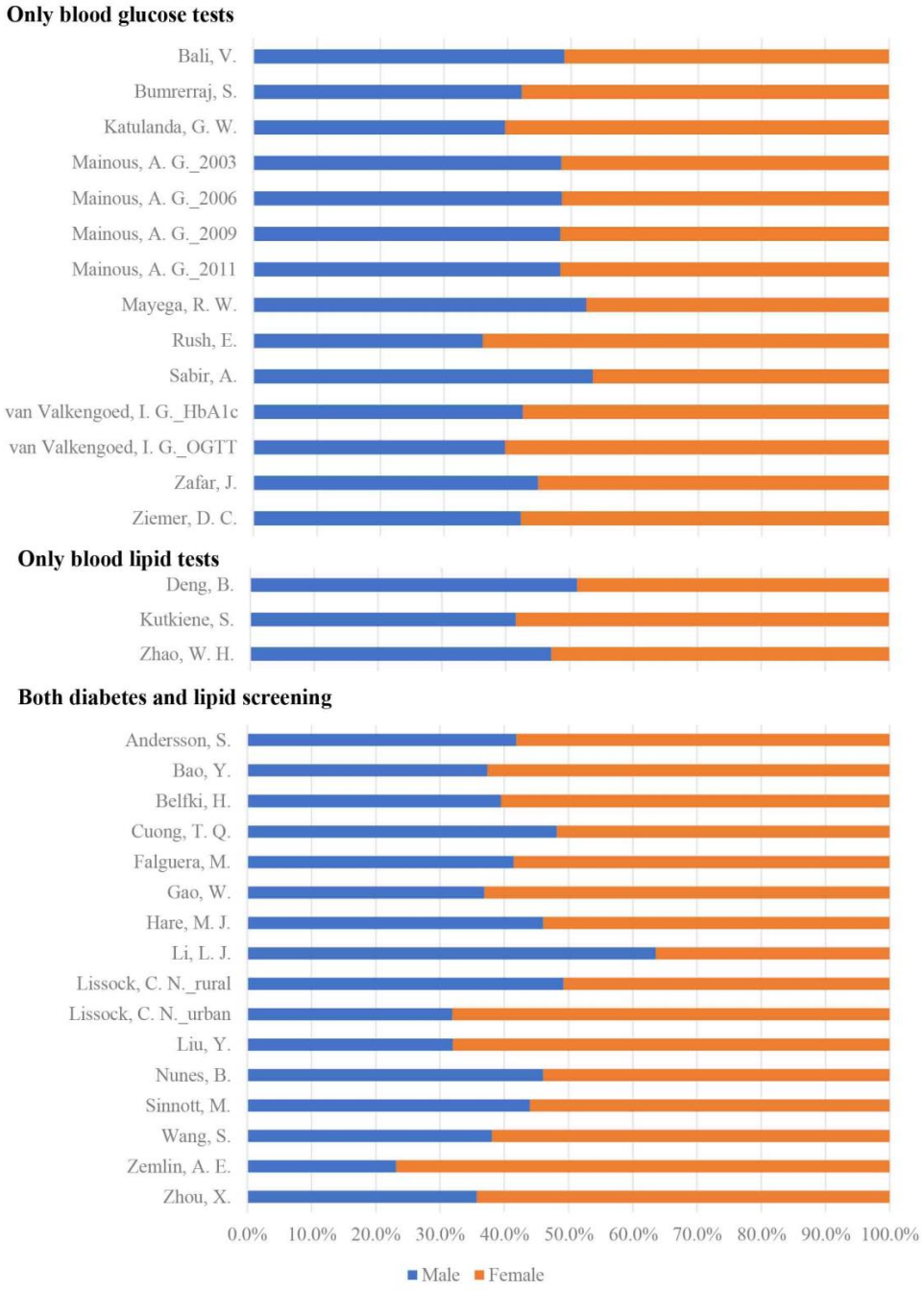
Supplementary table 3 Quality Assessment of included articles based on Appraisal of Cross-sectional Studies (AXIS)

Ding H, *et al*. *BMJ Open* 2022; 12:e055764. doi: 10.1136/bmjopen-2021-055764

Supplementary figure 1 Selection of articles for systematic review



Supplementary figure 2 The gender of screening participants



Supplementary table 1 Summary of diabetes and lipid screening guidelines

Country	Diabetes Screening Guidelines	Lipid Screening Guidelines
Australia	Patients should be screened for diabetes risk every three years from 40 years of age using the Australian type 2 diabetes risk assessment tool. Aboriginal and Torres Strait Islander peoples should be screened from 18 years of age. *(RACGP)*	Adults should have their blood lipids assessed every five years starting at 45 years of age. Aboriginal and Torres Strait Islander adults should have lipid tests performed every five years from 35 years of age. *(RACGP)*
Canada	Adults should be tested for type 2 diabetes every 3 years if they are age 40 years or over, or if they are at risk for T2DM. If they have risk factors for T2DM, they should be tested more frequently and/or start regular screening earlier. *(Canada Clinical Practice Guidelines Expert Committee)*	Initiating screening: In patients without CVD (primary prevention), we suggest lipid testing as part of global CVD risk estimation in men at age ≥40 years old and women at age ≥50 years old. Testing can be considered earlier for patients with known traditional CVD risk factors. *(College of Family Physicians of Canada)*
Singapore	Screening for T2DM should begin at 40 years and older; screening should also be considered in adults of any age who have one or more risk factors for diabetes. *(Singapore Ministry of Health)*	Clinicians should routinely screen men and women aged 40 years and older for lipid disorders, and younger adults for lipid disorders if they have other risk factors for coronary artery disease. *(Singapore Ministry of Health)*
United Kingdom	The UK National Screening Committee does not currently recommend screening for T2DM. This is because: there is no evidence that screening is more beneficial than not screening, and there is no evidence that finding T2DM early by screening results in a greater health benefit than usual health care.	National Institute for Health and Care Excellence guideline CG181 recommends (1.1.8) using the QRISK2 risk assessment tool (including ethnicity, age, sex, smoking, blood pressure, lipid profile, BMI, etc.) to assess CVD risk for the primary prevention of CVD in people up to and including age 84 years. There is also a potential need to amend recommendations 1.1.8 and 1.1.10 to advise the use of QRISK3 (including sex, age, BMI, lipid profile, blood pressure, family history, etc.) in place of QRISK2 because QRISK2 is due to be superseded by QRISK3 in 2018.
United States	Adults aged 40 to 70 who are overweight or obese are recommended to screen for abnormal blood glucose. (*USPSTF)*	Identification of dyslipidaemia and calculation of 10-year CVD event risk requires universal lipids screening in adults aged 40 to 75 years. *(USPSTF)*

RACGP: The Royal Australian College of General Practitioners, T2DM: type 2 diabetes mellitus, CVD: cardiovascular disease, USPTF: US preventive services task force recommendation statement

Supplementary table 2 Search strategy for article selection

Database	Search strategy	Result
PubMed	(("fasting plasma glucose" OR "postprandial plasma glucose" OR "oral glucose tolerance" OR "random plasma glucose" OR "glycated haemoglobin" OR "HbA1c" OR "A1C") AND (screening* OR test*) AND (uptake OR participate) NOT (Paediatric OR gestational))	**5,658**
Ovid Medline	(fasting plasma glucose OR postprandial plasma glucose OR oral glucose tolerance OR random plasma glucose OR glycated haemoglobin OR HbA1c OR A1C) AND (screening* OR test*) AND (uptake OR participate) NOT (Paediatric OR gestational)	**749**

Supplementary table 3 Quality Assessment of included articles based on Appraisal of Cross-sectional Studies (AXIS)

Author	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Ali, I.	1	1	1	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Andersson, S.	1	1	0	1	1	9	0	1	1	1	1	1	1	1	1	1	1	1	1	1
Bali, V.	1	1	0	1	9	0	0	1	1	0	1	1	0	0	1	1	1	1	9	9
Bao, Y.	1	1	0	1	0	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Belfki, H.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	9	1
Bumrerraj, S.	1	1	0	1	0	1	0	1	1	0	1	1	1	1	1	1	1	1	1	1
Cuong, T. Q.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	0	9	1
Deng, B.	1	1	1	1	1	1	1	1	1	1	1	1	0	1	1	1	1	0	1	1
Falguera, M.	1	1	1	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Gao, W.	1	1	0	1	1	1	0	1	1	0	1	1	0	0	1	1	1	0	9	1
Hare, M. J.	1	1	0	1	1	1	0	1	1	0	1	1	0	0	1	1	1	1	1	1
Hidalgo, L. A.	1	1	1	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Hwu, C. M.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Katulanda, G. W.	1	1	1	1	1	1	0	1	1	0	1	1	0	0	1	1	1	1	1	1
Koyama, A. K.	1	1	0	1	1	1	0	1	1	0	1	1	0	0	1	1	1	1	9	9
Kutkiene, S.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Li, L. J.	1	1	0	1	0	1	0	1	1	1	1	1	0	0	1	1	1	0	1	1
Lin, S.	1	1	0	1	1	0	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Lissock, C. N.	1	1	0	1	1	1	0	1	1	0	1	1	0	0	1	1	1	0	1	1
Liu, Y.	1	1	1	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Mainous, A. G.	1	1	1	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	9
Mayega, R. W.	1	1	1	1	1	0	1	1	1	0	1	1	0	1	1	1	1	1	1	1
Nunes, B.	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Rush, E.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	0	1
Sabir, A.	1	1	0	1	0	0	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Sinnott, M.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	0	9	1
Valkengoed, I. G.	1	1	0	1	1	1	1	1	1	1	1	1	0	1	1	1	1	1	1	1
Wang, S.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Zafar, J.	1	1	1	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Zemlin, A. E.	1	1	0	1	1	1	0	1	1	0	1	1	0	0	1	1	1	1	1	1
Zhao, W. H.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	0	1	1
Zhou, X.	1	1	0	1	1	1	0	1	1	1	1	1	0	0	1	1	1	1	1	1
Ziemer, D. C.	1	1	0	1	1	1	0	1	1	0	1	1	0	0	1	1	1	1	1	9

0: No, 1: Yes, 9: Don't know.